Higher National Diploma in Engineering (Civil/Mechanical)

Higher National Diploma in Building Services Engineering

First Year, First Semester Examination-2020

CE 1102: fluid mechanics

Answer any **five** questions. Time : 3 hours

Question 1

a) Explain the terms "**Total pressure**" and "**Centre of pressure**" related to a plate submerged in a static fluid. Write equations to find the **Total pressure** and **center of pressure** of an inclined plate submerged in a static fluid. Support your answer with clear sketches. (06 marks)

b) A $2\ \mathrm{m}$ **wide gate** supporting water is shown in Fig. 01. Find the **height** h **of the water** so that the gate tips about the hinge. (Hint: - Gate tips if resultant pressure force passes through a point in the range BC of the gate) . (14 marks)

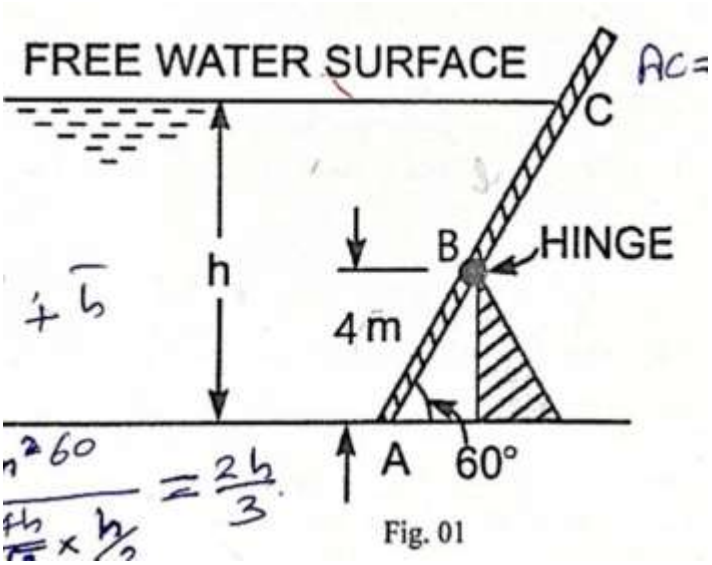

Fig. 01

Question 2

a) Explain what does it mean by "**viscosity**" in fluid mechanics. (02 marks)

b) What does it mean by "**Ideal Fluid**" & "**Real Fluid**"? (02 marks)

c) Two large plane surfaces are $2\ \mathrm{cm}$ **apart** and the space between the surfaces is filled with oil (**Fig. 02**). What **force (F) is required to drag a very thin plate** of surface area $0.5\ \mathrm{m}^2$ between the two large plane surfaces at a **speed of** $0.5\ \mathrm{m/s}$, if the thin plate is in the middle of the two plane surfaces. Take **dynamic viscosity of oil** $= 0.8\ \mathrm{Ns/m^2}$. (12 marks)

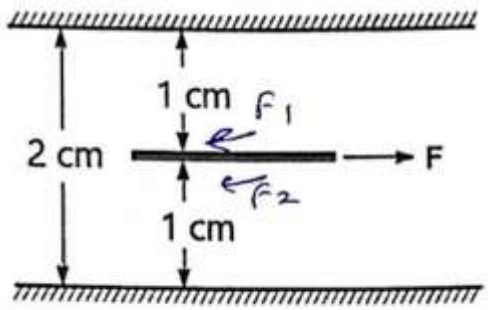

Fig. 02

d) Explain the operating principle of an **orifice type viscosity meter** with suitable sketches. (04 marks)

Question 3

a) Explain the **continuity equation**. (02 marks)

b) Explain **Bernoulli's equation**. (02 marks)

c) A **horizontal venturi meter** with inlet and throat diameters $32\ \mathrm{cm}$ **and** $16\ \mathrm{cm}$ respectively is used to measure the flow of water. The reading of the **differential manometer connected to the inlet and throat is** $18\ \mathrm{cm}$ **of mercury**. Determine the **rate of flow**. Take $C_d = 0.98$. (10 marks)

d) Describe the **operating principle of the Pitot tube** with suitable sketches and relevant equations. (06 marks)

Question 4

a) What is the **major cause of energy loss of fluid flow in pipes**? (01 marks)

b) Write **four minor causes of energy loss of fluid flow in pipes**. (03 marks)

c) The rate of flow of water through a horizontal pipe is $0.24\ \mathrm{m^3/s}$. The diameter of the pipe is $180\ \mathrm{mm}$ **is suddenly enlarged to** $300\ \mathrm{mm}$. The pressure intensity in the smaller pipe is $11.5\ \mathrm{N/cm^2}$. Determine: i) **loss of head due to sudden enlargement** ii) **pressure intensity in the large pipe** iii) **power lost due to enlargement** (06 marks)

d) **Three pipes** of lengths $750\ \mathrm{m}$, $500\ \mathrm{m}$ **and** $420\ \mathrm{m}$ and diameters $500\ \mathrm{mm}$, $450\ \mathrm{mm}$ **and** $350\ \mathrm{mm}$ respectively are connected **in series**. These pipes are to be replaced by a **single pipe of length** $1600\ \mathrm{m}$. Find the **diameter of the single pipe**. (04 marks)

Question 5

a) What does it mean by **drag force**? (02 marks)

b) What are the **two types of drag forces**? (02 marks)

c) With clear sketches explain a **streamlined body** and a **bluff body**. (04 marks)

d) A partially submerged body is towed in water. The **resistance** R **to its motion** depends on the **density** ρ, **viscosity** μ of the water, **length** L of the body, **velocity** V of the body and the **acceleration due to gravity** g. Show that the resistance to the motion can be expressed in the form:

$$R = \rho L^2 V^2 \phi \left[\left(\frac{\mu}{\rho V L} \right) \cdot \left(\frac{Lg}{V^2} \right) \right]$$

(12 marks)

Question 6

a) A **jet of water of diameter** $80\ \mathrm{mm}$ moving with a **velocity of** $30\ \mathrm{m/s}$ strikes a **fixed plate** in such a way that the **angle between the jet and the plate is** 60°. Find the **force exerted by the jet on the plate**: i) in the **direction normal to the plate** ii) in the **direction of the jet** (08 marks)

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b) Draw a clear sketch and name the main parts of a **centrifugal pump**. (04 marks)

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c) Draw a clear sketch and name the main parts of a **single-acting reciprocating pump**. (04 marks)

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d) Draw a clear sketch and name the main parts of a **Pelton turbine**. (04 marks)